

CLINICAL MULTIPLE CHOICE EVALUATION

SEMINAR 5 OF 20: SKIN CONDITIONS AND WOUND HEALING



Candidate Instructions

- This textbook-derived examination covers **25 Single Best Answer (SBA)** multiple-choice items.
- All items map directly to the anatomical, structural, pathological, and healing criteria matching your class seminar.
- Select the single best choice (**A, B, C, or D**) for each question item.
- The absolute, comprehensive evaluation key matrix block is attached at the end of this paper.

QUESTION 1

The outer part of the skin is known as the epidermis and consists of four strata representing stages in cell maturation. What is the correct cell progression direction?

- A) From an outer corneal layer inward to a germinal layer
- B) From a germinal layer outward to an outermost corneal layer
- C) From the reticular dermis to the papillary dermal junction
- D) From a dense subcutaneous fat cluster into the spiny stratum spinosum

QUESTION 2

Match the specific epidermal layers with their core architectural characteristics as stated in the slides. Which statement is perfectly accurate?

- A) The stratum corneum is composed of large, actively dividing germinal cells
- B) The basal layer contains cells with lost nuclei that are flattened and packed with keratin
- C) The stratum spinosum, or spiny layer, features spikey protrusions that help anchor cells together
- D) The stratum granulosum features high fluid content and expanding cytoplasm thickness

QUESTION 3

The epidermis houses specialized cellular elements. Which specific cells are explicitly defined as being "involved in immune function"?

- A) Melanocytes
- B) Langerhans cells
- C) Merkel cells
- D) Fibroblasts

QUESTION 4

What are Merkel cells explicitly responsible for within the epidermal layers?

- A) Synthesis of oily sebum secretions
- B) Melanin manufacturing to block UV light lines
- C) Sensory perception regarding light touch
- D) Consumption of structural keratin inside the hair matrices

QUESTION 5

According to the clinical rule outlined on Slide 9, what happens if the basal layer of the epidermis is completely destroyed?

- A) The skin will experience an immediate transformation into thick hairless type blocks
- B) The skin will not be able to regenerate, requiring a skin graft (autograft)
- C) The apocrine glands accelerate production of high-protein fluids
- D) The tissue proceeds immediately into maturation phase type 1 collagen synthesis

QUESTION 6

The dermis is divided into the papillary dermis and the reticular dermis. What specific clinical effect occurs when the dermis suffers tears under extreme stretching?

- A) The manifestation of tinea pedis fungal tracks
- B) The production of striae or stretch marks
- C) The complete loss of local eccrine sweat output capabilities
- D) Malignant degeneration of benign nevi components

QUESTION 7

Blonde or red hair variants exhibit different tracking traits compared to dark hair. What chemical components are structurally increased in blonde or red hair variants?

- A) Nitrogenous waste and lactic acid
- B) More sulphur and iron within the melanin variant
- C) High-density liquid lipids and specialized enzymes
- D) Type III collagen strings exclusively

QUESTION 8

Which skin glands secrete oily sebum that moistens the skin, helps prevent hair from drying out, and can lead to acne if inflamed?

- A) Eccrine sweat glands
- B) Apocrine sweat glands
- C) Ceruminous glands
- D) Sebaceous glands

QUESTION 9

What is the explicit operational difference between eccrine and apocrine sweat glands?

- A) Eccrine glands produce thick earwax solutions, while apocrine glands manage lactation lines
- B) Eccrine glands are the most common and secrete water with ions for cooling; apocrine glands produce a more viscous solution with protein/lipids stimulated by emotional factors
- C) Eccrine glands are restricted to palms and soles, whereas apocrine glands pack the entire body surface area
- D) Eccrine glands respond to emotional strain, while apocrine glands respond exclusively to environmental heat changes

QUESTION 10

The slides describe 5 key functions of the skin. Which process describes vitamin D production?

- A) It is excreted via eccrine waste streams in combination with glucose and lactic acid
- B) It is absorbed directly via thin hairy skin matrices from the atmosphere
- C) It is manufactured from cholesterol in response to UV radiation
- D) It is compiled by phagocytic macrophages during the acute inflammatory phase

QUESTION 11

Review your seminar lesion types. Which definition accurately matches a "Macule"?

- A) An elevated rough lesion more than 1cm in diameter (e.g., psoriasis)
- B) A flat area with a change of colour (e.g., freckles, nevi, measles)
- C) An elevated firm area less than 1cm in diameter (e.g., wart, insect bite)
- D) A profound loss of epidermis and dermis leading to tissue necrotization

QUESTION 12

Vitiligo and port-wine stains are explicitly classified as which specific type of primary skin lesion?

- A) Plaque
- B) Patch (a flat, irregular-shaped macule more than 1cm in diameter)
- C) Papule
- D) Ulcer

QUESTION 13

What is the underlying physiological mechanism that initiates the production of a decubitus (pressure) ulcer?

- A) Malignant proliferation of epidermal keratinizing squamous epithelial cells
- B) Ischaemia (reduced blood flow) triggered by unrelieved pressure, shearing forces, friction, or moisture leading to tissue necrosis
- C) T-helper cell hyper-activation prompting hyperpigmentation boundaries
- D) Fungal digestive enzymes consuming the basement membrane zones

QUESTION 14

Why does the pressure ulcer slide state that initial damage may be harder to assess in individuals with darker skin?

- A) Because darker skin contains thick hairless layers across all body quadrants
- B) Because the initial ischaemic skin damage indicators may be harder to visually assess
- C) Because melanocytes stop producing melanin over bone prominences entirely
- D) Because skin stretching causes an immediate conversion of type I collagen into type III

QUESTION 15

Pruritus (itching) is highly prevalent across dermatology. Which systemic conditions are explicitly noted as being capable of leading to pruritus?

- A) Arsenic poisoning, striae development, and autograft failure
- B) Renal failure, cholestatic liver disease, thyroid disorders, and drug reactions
- C) Squamous cell carcinoma, melanoma degeneration, and staphylococcus clusters
- D) Angiogenesis collapse, type I scar tissue mutations, and macular atrophy

QUESTION 16

Psoriasis is defined as a chronic relapsing, proliferative, inflammatory skin disorder. What classification and core cell mechanism define this disease?

- A) An allergic hypersensitivity reaction involving sensitized T-cells reacting to external metal toxins
- B) An autoimmune disease based on T-helper activation, leading to thickened epidermis/dermis with proliferation of skin cells
- C) A bacterial infiltration of connected hair follicles leading to systemic carbuncles
- D) A surface epithelial tumor originating directly from mutated germinative Merkel lines

QUESTION 17

Bacterial infections can target the skin structures. What specific condition manifests if an infection of staphylococcus aureus or streptococci results in infection of the dermis and subcutaneous tissue?

- A) Candidiasis
- B) Cellulitis
- C) Tinea Capitis
- D) Psoriasis Plaque tracks

QUESTION 18

Why is the skin so prone to fungal infections like Tinea, and how do they cause localized inflammation?

- A) Fungi invade the rich dermal blood vessels and consume white blood cells
- B) Fungi consume keratin and cause inflammation from their digestive enzymes
- C) Fungi transform eccrine water sweat outputs into thick lipids
- D) Fungi mutate tumor-suppressing genes inside the basal layer strata

QUESTION 19

Regarding the three main types of skin cancer, which of the following perfectly matches the description of Basal Cell Carcinoma?

- A) It is the least common type but the most common cause of death from skin cancer
- B) It originates from basal/germinative cells; metastatic spread is rare because it does not invade blood vessels or lymphatics
- C) It is highly associated with drinking water containing massive levels of arsenic in the UK
- D) It develops exclusively from pre-existing type I scar tissue configurations

QUESTION 20

Which skin cancer is defined as a malignant tumor of the cells that stimulate melanin production, is the least common type, but the most common cause of death from skin cancer?

- A) Squamous Cell Carcinoma
- B) Basal Cell Carcinoma
- C) Cutaneous Melanoma
- D) Pilosebaceous Sarcoma

QUESTION 21

According to the seminar presentation, what are the exact criteria elements for the ABCDE rule used in the clinical assessment of melanoma?

- A) Absence, Boundary, Chronicity, Depth, Exudate
- B) Asymmetry, Border irregularity, Colour variation, Diameter larger than 6mm, Elevation
- C) Autograft, Basal tier, Cellulitis, Dermatitis, Eccrine flow
- D) Abrasion, Blistering, Capillary count, Dermal depth, Elastin matrix

QUESTION 22

During the dynamic phases of wound healing, what physiological events characterize the "Inflammatory Phase"?

- A) The rapid development of new blood vessels via angiogenesis loops to supply oxygen
- B) Initial vessel contraction and clot formation, followed by vasodilation allowing white blood cells/enzymes to reach the area, featuring neutrophils and macrophages
- C) The active remodeling of collagen from flexible type III strands into firm type I strings
- D) The immediate resurfacing of wound borders via specialized epithelialisation cells

QUESTION 23

Under the "Proliferation" phase slide, what clinical indicator is highlighted regarding the appearance of granulation tissue?

- A) Healthy tissue is dark purple, completely smooth, and bleeds with minimal movement
- B) It is granular and uneven in texture, pink/red in colour, and does not bleed easily; dark granulation can indicate poor perfusion, ischaemia, and/or infection
- C) It is composed entirely of dead keratinizing epithelial scales that block bacterial entry
- D) Cellular activity completely halts and the aggregate number of blood vessels drops to zero

QUESTION 24

What specific structural modification occurs during the "Maturation Phase" of wound healing once the wound has closed?

- A) Macrophage cell counts double to execute debridement tasks
- B) Remodelling of collagen from type III (flexible) to type I (associated with scar tissue)
- C) Sutures or staples are automatically metabolized by epidermal digestive enzymes
- D) Angiogenesis accelerates to quadruple the number of local capillaries

QUESTION 25

Which healing category describes a scenario where "there is no total penetration of the dermis, wound edges are brought together so they are adjacent, and closure is performed with sutures, staples, or tape"?

- A) Secondary intention healing
- B) Delayed primary closure or tertiary suturing
- C) Primary intention healing
- D) Tissue autograft reconstruction closure

OFFICIAL EXAM ANSWER KEY MATRIX

Verify student choice selections using this unalterable evaluation matrix block.

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10
B	C	B	C	B	B	B	B	B	C

Q11	Q12	Q13	Q14	Q15	Q16	Q17	Q18	Q19	Q20
B	B	B	B	B	B	B	B	B	C

Q21	Q22	Q23	Q24	Q25
B	B	B	B	C